

Chapter 19

- 1. Explain the role of intertemporal substitution of labor in the real business cycle theory. Based upon this theory, would you expect the real wage to be procyclical or countercyclical? (Hint: Think of labor productivity in the Robinson Crusoe example and the link between marginal productivity and the real wage.)**

Answer:

1. In real business cycle theory, the intertemporal substitution of labor plays the primary role in determining labor supply. People are willing to work harder only if they receive a higher intertemporal real wage. Based upon this theory, an economic boom that includes higher employment must mean that the workers are receiving a higher real wage. The higher wage is the incentive needed to boost employment. Therefore, this theory would predict that wages are procyclical.

- 2. How does labor hoarding affect the Solow residual in a recession?**

Answer:

2. The Solow residual is the percentage change in output minus the percentage change in inputs weighted by factor shares. In a recession, output has fallen, but if firms hoard their workers, there will not be an accompanying fall in employment. Therefore, the Solow residual will appear to fall by a larger degree due to the labor hoarding.

- 3. How does the presence of an aggregate-demand externality affect prices and output in the economy?**

Answer:

3. An aggregate-demand externality of the variety seen in menu cost studies leads to higher prices and lower output. This is caused by the fact that an individual firm has little incentive to change prices frequently due to the menu cost it must pay for each change. For example, if a firm has set its price too high,

it may not want to incur the menu cost to reduce the price. Society would prefer that the firm would lower its price because this would lead to a boost in sales (and output). If there are many such firms that behave this way, the externality for society becomes large.

4. Is money neutral in the short run in a model of staggered wage and price setting? Is it neutral in the long run? Explain.

Answer:

4. In the staggered wage and price setting model, unexpected changes in the money supply will lead to output effects in the short run because firms are unable to adjust prices and wages. In the long run, however, firms will eventually be able to adjust to the changes in the money supply. Therefore, this model predicts that money is neutral in the long run, but not in the short run.

Chapter 18

1. Explain how the Baumol-Tobin model supports the specification of the money demand function, $L(i,y)$.

1. The Baumol-Tobin model provides a framework in which an increase in the interest rate lead to a reduction in money demand (average money holdings) and an increase in output leads to an increase in money demand.

2. Suppose the banks in an economy have a reserve-deposit ratio of 10 percent and the currency-deposit ratio is 20 percent.

a. If the Federal Reserve increases the monetary base by \$500 through open market operations, what will be the increase in the money supply?

b. If the Federal Reserve increases the discount rate and firms react by increasing the reserve-deposit ratio to 15 percent, what is the change in the multiplier? Will this change increase or decrease the money supply?

c. In anticipation of the Y2K problem, many people withdraw their deposits from the bank in December 1999. What impact will this have on the money multiplier, high powered money, and the money supply?

Answers:

- 2. a.** The money multiplier is $(0.2 + 1)/(0.2 + 0.1) = 4$. An increase in the monetary base of \$500 will increase the money supply by \$2000 (500×4).
- b.** The new money multiplier is $(0.2 + 1)/(0.2 + 0.15) = 3.43$. This will lead to a decrease in the money supply because banks will hold more reserves and issue less loans.
- c.** If people withdraw their deposits, this will increase the currency-deposit ratio. This in turn will decrease the money multiplier. High powered money will decrease due to the fall in reserves caused by the decrease in deposits. The money supply will decrease due to both the reduction in the multiplier and the decrease in high powered money.

3. Explain how changes in the money multiplier contributed to the severity of the Great Depression. What could the Federal Reserve have done to restore the money multiplier to its original level?

Answer:

3. The primary change in the money multiplier was due to the large number of bank failures. This led to an increase in the currency-deposit ratio due the drop in confidence in the banking system. The increase in the ratio led to a fall in the money multiplier and in the money supply. The large number of bank failures led the remaining banks to increase the reserve-deposit ratio, which further decreased the money multiplier and the money supply. The actions in response to the banking crisis lead to a huge fall in the money supply which in turn led to high interest rates and a fall in investment. To restore the money multiplier to its original level, the Federal Reserve would have needed to address the public's fears of the banking system by providing emergency loans and additional money to the system.

4. In the Baumol-Tobin model, trips to the bank cost \$10. The interest rate is 10 percent.
- a. If Paul plans to gradually spend \$1800 this year, how many trips to the bank should Paul make? What is his average money holding?
- b. If interest rates increase to 15 percent, how does this affect the number of trips Paul will make to the bank?
- c. If Paul moves further away from the bank, so that trips now cost \$20, what will be the change in his average money holdings?

Answers:

4. a. The number of trips to the bank is solved as

$$N = \text{square root of } 0.1 \times 1800 / (2 \times 10) = 3 \text{ trips to the bank}$$

$$\text{The average money holding is } 1800 / (2 \times 3) = 300.$$

- b. The number of trips increases to 3.67 on average (square root of $0.15 \times 1800 / (2 \times 10)$).

- c. The increase in the fixed cost of a visit to the bank will increase average money holdings to \$424 (square root of $1800 \times 20 / (2 \times 0.1)$).

5. How has the introduction of near money changed the Federal Reserve's ability to control the money supply? Do you think this will diminish the power of the Federal Reserve? Explain.

Answer:

5. The introduction of near money has made the demand for money unstable as people now have more alternative ways of holding their liquid assets. The Federal Reserve has reacted to this problem by placing emphasis on the setting of interest rates, which it can control more accurately. Thus far, the Federal Reserve has proved to have maintained its effectiveness in the implementation of monetary policy.

Chapter 17

1. Explain how the following events affect the rental price of capital in an economy with a Cobb-Douglas production function:

- a. A large earthquake destroys 20% of the capital stock.**
- b. A supercomputer is invented that increases the productivity of each worker.**
- c. The labor force shrinks as the baby boom generation begins to retire.**

Answers:

- 1. a.** A decrease in the capital stock will increase the marginal productivity of the remaining capital. This will cause the rental price to increase.
- b.** An increase in technology will increase the marginal productivity of the remaining capital. The rental price will increase.
- c.** A decrease in the labor force will lower the marginal productivity of capital, and the rental price will decrease.

2. How is the stock market a good indicator of economic activity? Does an increase in the stock market signal a good time to increase or decrease the capital stock?

Answer:

2. Based upon Tobin's q theory, a change in the stock price will alter the value of Tobin's q . If the stock market rises, this increases the value of q , making investment more attractive. The stock market also is an important part of household wealth. An increase in the stock market could lead to increased consumption due to increased wealth of consumers. Finally, a change in the stock market may reflect changes in expectations of future profits, thus signaling changes in the natural rate of output.

3. The federal government offers loans to small business that would otherwise be unable to borrow funds.

a. How does this program affect output in the economy?

b. What is the danger in making these loans? Could this program actually cause a decrease in output?

Answers:

3. a. This government program helps firms that face financing constraints. By providing loans, small firms can undertake profitable investments. This will lead to a boost in GDP.

b. The danger in this program is that some of the loans may go to unprofitable investments. If there are too many bad loans, this extra government spending may lead to a decrease in GDP because the money could have been put to better use elsewhere.

4. How will the following events affect the supply of new residential housing:

a. an end of rent control for apartments

b. a decrease in the real interest rate

c. an end of the tax deduction for mortgage interest

d. a decrease in the population

Answers:

4. a. An end of rent control will increase the supply of new housing because the higher costs of apartments will increase housing demand.

b. A decrease in the real interest rate will increase the supply of housing due to the lower cost of borrowing and the lower opportunity cost of housing investments.

c. An end to the mortgage interest deduction will lower the supply of housing. The cost of owning a home has increased due to the loss of the tax deduction.

d. A decrease in population will lower the housing supply due to the accompanying decrease in housing demand.

5. Compare the production smoothing motive for holding inventories with the accelerator model. Do these models predict the same pattern of inventory behavior as the economy enters a predicted economic boom? How would your answer differ if the boom was unpredicted?

Answer:

5. The production smoothing model predicts that firms will build up inventories during times of low sales and reduce inventories during periods of high sales.

The accelerator model predicts that firms will increase inventories during periods of high output. In a predicted economic boom, firms would anticipate the increase in sales. According to the production smoothing model, firms would have built up inventories before hand and allow the stocks to diminish during the boom. The accelerator model predicts that if firms expect a boom and increase production, they will increase their inventory holdings. If the boom is unexpected, inventories will still fall in the production smoothing model. In the accelerator model, firms will not have increased production because they do not expect a boom. Therefore, inventories will not increase, and they may actually decrease due to the high demand for goods.

Chapter 16

1. Why did the Keynesian consumption function lead to fears of secular stagnation?

1. The Keynesian consumption function predicts that as income grows over time, people will consume a smaller share of this additional income. The fear was that people would save too much and there would be no good investment opportunities for which to use the funds. If this process continued, eventually the economy would stagnate.

2. In the two-period Fisher model, suppose Bill earns \$100 in the first period and \$200 in the second period.

a. If Bill consumes \$140 in the first period and \$150 in the second period, what is the interest rate?

b. Now if Bill's consumption changes to \$150 in the first period and \$140 in the second period, what is the new interest rate?

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- c. Explain the income and substitution effect and discuss which effect dominated after the interest rate change from part (a) to part (b).
- d. Based upon Bill's actions, is he consumption smoothing?

Answer:

2. a. The interest rate is determined by solving the following equation for r:

$$140 + 150/(1+r) = 100 + 200/(1+r)$$

$$40 = 50/(1+r)$$

$$1+r = 1.25$$

$$r = 0.25 \text{ or } 25 \text{ percent}$$

- b. The new interest rate is solved as:

$$150 + 140/(1+r) = 100 + 200/(1+r)$$

$$50 = 60/(1+r)$$

$$1+r = 1.2$$

$$r = .2 \text{ or } 20 \text{ percent}$$

- c. As the interest rate decreases, this will cause the budget constraint to rotate. In part (a), Bill was borrowing money in the first period to finance consumption. Now that the interest rate has fallen, it has lowered the cost of borrowing. The income effect is the change in consumption caused by the movement to a higher indifference curve. Since borrowing costs are lower, Bill will feel that he has a higher real income. Therefore, the income effect says that he will consume more of each good. The substitution effect is the change in consumption resulting from the change in the relative price of consumption in each period. Since it is now less expensive to borrow to finance consumption in the first period, the substitution effect says that Bill will consume more in the first period than before and less in the second period. To determine which effect dominated, examine the overall change in consumption. For consumption in the first period, both the income and substitution effects state that he should consume more. For consumption in the second period, the income effect states that he should

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consume more, but the substitution effect says the reverse. Bill's consumption fell from 150 to 140, so the substitution effect dominated.

d. Yes, Bill is consumption smoothing because he is borrowing in the first period against his higher income in the second period.

3. What is the difference between Modigliani's life-cycle hypothesis and the Keynesian consumption function?

Answer:

3. The Keynesian consumption function states that consumption is only affected by current income. The life-cycle hypothesis states that consumption is based upon a person's lifetime resources, composed of wealth and lifetime earnings.

4. Based upon the permanent income hypothesis, how would you expect consumption today to be affected by the following changes:

- a.** A worker takes a two-year leave from a job to enroll in business school.
- b.** A person decides to accept an early retirement offer. Before accepting the offer, the worker had planned on working 10 more years. As part of the offer, the company offered the worker 80 percent of the final salary over those 10 years, and the benefits for the following years would remain unchanged.
- c.** The government issues a temporary tax increase.
- d.** A worker fails to get a 20 percent raise that he had counted upon receiving.

Answer:

4. a. A business school degree should significantly increase lifetime earnings, despite the cost of going to school. Therefore, the person should increase consumption today.

b. Based upon the early retirement offer, the expected income has permanently fallen. This should lead to a decrease in consumption.

c. A temporary tax increase should have very little effect on consumption.

d. Even though actual income did not fall, the worker will lower consumption because his expected permanent income has decreased.

5. In the permanent income model, how do borrowing constraints alter the effect of a temporary increase in taxes?

Answer:

5. In the permanent income model, a temporary increase in taxes should have very little effect on output, but if consumers face borrowing constraints the results will differ. Normally, people would simply borrow money to pay for the temporarily higher taxes and allow consumption to remain unchanged. But if they cannot borrow funds, they will have to lower consumption in order to pay the higher tax bill. This decrease in consumption will lead to a decrease in output.

Chapter 15

1. Supposed that the government currently has a balanced budget. The politicians note that consumers spend 80 percent of their disposable income.

a. What is the tax multiplier?

b. If the policy makers use this multiplier, how much of a tax reduction is required to boost GDP by \$100 million?

c. According to Ricardian equivalence, do you think the government estimate is correct? Why or why not?

Answers:

1. a. The tax multiplier is $-MPC/(1-MPC) = -.8/.2 = -4$.

b. To boost GDP by \$100 million, the government needs to cut taxes by \$25 million ($100/4$).

c. According to Ricardian equivalence, people will save the tax cut because they know that the government is going to have to repay the deficit later. Thus, there will be no boost to consumption or GDP.

2. a. Explain the link between life-expectancy and forecasts of future government deficits.

b. How would these forecasts be affected by:

i. increases in the retirement age

ii. increases in immigration

iii. increases in birth rates

Answers:

2. a. As life expectancy increases, people will be receiving Social Security payments for a longer period of time. Thus, the government expenditures will increase with the larger and longer-lived elderly population, but tax revenues will not increase since the working-age population has not changed. As a result, government deficits will become larger.

b. i. Increases in the retirement age will reduce forecasted deficits because people will be earning income (and paying income taxes) for a longer period, and they will receive Social Security benefits for a short period.

ii. Increases in immigration could have various effects. If the immigrants are of working age, they will help reduce the deficit through their income taxes. If the immigrants are too young or old to work, they will lead to an increased deficit through increased government expenditures on services such as schooling and medical aid.

iii. Increases in birth rates will initially increase the deficit through costs of public education. As this group reaches working age, they will help reduce the deficit through income taxes and contributions to Social Security.

3. In 1999, the United States government had a large budget surplus. Some politicians called for tax cuts to return this surplus to the people.
- a. How do you think a cyclically-adjusted budget surplus would differ from the reported surplus? If the tax cuts were made, what would happen to the cyclically-adjusted budget balance if the economy went into a recession?
- b. How would the inclusion of uncounted liabilities affect the budget surplus?

Answers:

3. a. In 1999, the economy was in an economic boom and the government was receiving large tax revenues from the high output. A cyclically-adjustment budget measurement would show a much smaller surplus (or possibly a deficit). If tax cuts were made, the cyclically-adjusted budget measurement would fall. In a recession, this measurement would not change much because the measurement is based upon assuming output is near the full-employment level. The regular budget measurement, on the other hand, would display a huge deficit as tax revenues fell due to lower output/income.

b. The inclusion of uncounted liabilities would turn the budget surplus into a deficit and increase the government debt because of the large population that is nearing retirement. The government has committed to paying retirees Social Security and Medicare, so the large liabilities would outweigh the contributions of a smaller future workforce.

4. What are the benefits of inflation-indexed bonds? How can these bonds assist an inflation-targeting central bank?

Answers:

4. Inflation-indexed bonds provide the opportunity for people to invest at a guaranteed real interest rate, since the nominal return adjusts with inflation. These bonds also reduce the government's incentive to increase inflation as a way of lowering the real value of repayment. For central banks, these bonds help predict the market's expected inflation. By monitoring the trading price for these assets, the central bank can infer the predicted inflation by the market. The central bank can then alter policy if inflation expectations move outside of the target zone.

Chapter 14

1. According to the Lucas critique, why should policy makers not rely upon the Phillip's curve relationship between inflation and unemployment as the formula for monetary policy.

Answer:

1. If policy makers consistently used the Phillip's curve to determine how much inflation was needed to lower unemployment, people would react to this policy by expecting higher inflation every time unemployment was high. This change in inflation expectations would cause an outward shift in the Phillip's curve, and an even higher rate of inflation would now be needed to lower unemployment.

2. Describe how automatic stabilizers can help to "cool off" an economic boom.

Answer:

2. Automatic stabilizers are policies that help stimulate or slow the economy when needed without requiring any specific policy change. One stabilizer that helps slow a booming economy is the income tax. As people earn more in a boom, the amount of taxes they pay increases. This prevents consumers from spending all of their additional income, which would further boost the economy.

3. Describe two benefits and two costs of an inflation-targeting rule for monetary authorities.

Answer:

3. The benefits of inflation targeting include the increased accountability of the central bank because people are able to easily judge its performance. Another benefit is added credibility caused by providing clear guidelines for the goals of monetary policy. Costs of inflation targeting include a lack of discretion to address other economic problems that may occur. Also, the targeting may require extreme policy adjustments to keep inflation within the required bands, which may lead to higher volatility in other areas such as GDP and unemployment.

4. Why do you think the level central-bank independence is not correlated with a country's average growth rate of real GDP?

Answer:

4. From our study of money neutrality in previous chapters, we know that monetary policy does not affect real GDP in the long run. Thus, we should expect that the degree of central-bank independence is not correlated with the country's average (long run) growth rate

Chapter 13

1. Answer the following questions about the short-run aggregate supply equation:

$$Y = \bar{Y} + \alpha(P - P^e)$$

a. Let $\alpha = 0.5$. Draw the aggregate supply curve.

b. As the term α increases, how does this change the supply curve?

c. If the expected price level increases, how does this change the supply curve?

Answers:

1. a. The aggregate supply curve should be graphed from the following equation:

$$P = \left(P^e - \frac{1}{\alpha} \bar{P} \right) + \frac{1}{\alpha} \bar{P}$$

where the first group of terms are the intercept and the slope is $1/\alpha$.

b. As α increases, the aggregate supply curve becomes flatter and the intercept with the y-axis increases

c. An increase in the expected price level will shift up the supply curve.

2. The sticky-wage model is based upon the assumption that nominal wages are designated by long-term contracts.

a. If the variability of inflation increases, how would this affect the desired contract length of workers? If the contract length was changed, how would this affect the calculation of the expected price level -- would it change faster or slower in response to a price change?

b. If workers stopped signing long term contracts, how would this change the aggregate supply curve?

Answers:

2. a. As the variability of inflation increases, workers will want shorter contract lengths since inflation is more unpredictable. The shorter contracts essentially allow workers to update their price expectations more frequently with the signing of new contracts. Thus, the expected price level should respond more quickly to a price change.

b. If workers no longer signed long term contracts, wages would be as flexible as prices. The aggregate supply curve would now be a vertical line.

3. What is the difference between cost-push and demand-pull inflation? Which was the primary cause of inflation in the early 1970's? What type of inflation has the Federal Reserve been trying to prevent in 1998 and 1999?

Answer:

3. Cost-push inflation is caused by supply shocks that alter the cost of production. Demand-pull inflation is caused by upward pressure on prices from high aggregate demand. In the early 1970's, the OPEC oil price shock was a primary cause of the high inflation. This is an example of cost-push inflation. In 1998 and 1999 the Federal Reserve was worried about demand-pull inflation caused by high demand in the current economic boom.

4. a. Explain what it means for real wages to behave cyclically.

b. Based upon this cyclicity, do you think workers are more productive or less productive in an economic boom? (Hint: Link the real wage with marginal productivity of labor.)

c. If an economic boom was caused simply by firms hiring more workers, would you expect the real wage to move cyclically? What could explain the cyclicity of real wages?

Answers:

4. a. Cyclicity of wages means that real wages increase as real GDP increases.

b. In competitive models, firms will only pay a higher wage if workers are more productive. Thus, the productivity of the last worker hired (marginal productivity) has increased when we view a higher real wage.

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c. If output was increased by just hiring more workers, we would expect the real wage to fall due to diminishing marginal productivity. Without adding more capital and other input factors, each additional worker will be less productive. In this situation, we would expect the real wage to fall. The cyclical nature of real wages must be caused by other factors, such as sticky prices or technology shocks that shift the labor demand curve. An outward shift in the labor demand curve will increase employment and the real wage.

5. a. Explain the link between the aggregate supply curve and the Phillip's curve.
- b. If the natural rate of unemployment falls, how will this affect the Phillip's curve?
- c. If the sacrifice ratio was 5 and the central bank wanted to lower inflation by 8 percentage points, how much GDP would have to be sacrificed? Would you recommend reducing the inflation all at once, or spreading it out over several years? What would a proponent of rational expectations theory recommend?

Answers:

5. a. The aggregate supply curve provides the link between output and prices, which can be modified to link output and inflation. By incorporating Okun's law, which links output and unemployment, we then have the Phillip's curve relationship between inflation and unemployment.

b. The Phillip's curve is graphed from the following equation:

$$\pi = (\pi^e + v + \beta u^n) - \beta u$$

where the first group of terms is the y-axis intercept and the slope is $-\beta$. A decrease in the natural rate will shift down the Phillip's curve.

c. The sacrifice ratio is the percentage of a year's GDP that must be forgone to reduce inflation by 1 percentage point. With a sacrifice ratio of 5, lowering inflation by 8 percentage points would require 40 percent of GDP to be sacrificed. This would be a lot of GDP to lose in a one-year period, so it may be best to spread it out over several years in order to avoid mass unemployment. On the other hand, a proponent of rational expectation would argue that committing to

the policy with a quick reduction would be the best solution. They argue that this quick-reduction policy will alter people's expectations and allow prices to fall quickly without the large loss in GDP.

Chapter 12

1. A hurricane has just wiped out several shipbuilding companies in a small open-economy country. These companies were the primary exporters for the country. Assume that the country had been at full employment. In addition, the country has a flexible exchange rate.

a. If the government does nothing to respond to this crisis, what will happen to output, the trade balance, and the exchange rate in the short run?

b. Should the government intervene in this situation? If so, what policy would you recommend?

c. How would your answers have changed if the country had a fixed exchange rate?

Answers:

1. a. The reduction in the export industry will lead to an inward shift in the net exports schedule. This will cause an inward shift in the IS^* schedule. Since the LM^* curve is vertical, output will remain unchanged, but the exchange rate will fall. This will serve to reduce imports to match the export reduction caused by the hurricane. The trade balance will therefore remain unchanged.

b. With a floating exchange rate, the government does not need to intervene. Output will remain at its original level.

c. With a fixed exchange rate, output would fall as a result of the inward shift in the IS^* curve. The government should respond with increased government spending to make up for the decrease in exports until the shipbuilding industry has time to rebuild.

2. Suppose a small country's risk premium decreased due to a reduction in the government deficit. Assume that the country has a flexible exchange rate.

a. Show the effect of this change on the IS*-LM* graph. What has happened to output and the exchange rate?

b. What has happened to the trade balance?

Answers:

2. a. A decrease in the risk premium will lower the interest rate and shift out the IS* curve due to higher investment. The LM curve will shift inward due to the increase in money demand caused by the lower interest rate. As a result, output has decreased and the exchange rate has increased.

b. The increase in the exchange rate will cause the trade balance to worsen as imports become cheaper and exports more expensive.

3 Explain why the LM* curve is vertical.

Answer:

3. The LM* curve is vertical because the interest rate must match the world interest rate. This means that the output level is determined in the money market, independent of any exchange rate. Thus, the LM* curve is vertical to reflect that the exchange rate does not influence output.

4. Describe changes in aggregate output, the exchange rate, and the trade balance in response to the following effects on the country under a fixed exchange rate system:

- a. an increase in taxation
- b. a decrease in investor confidence that leads to a reduction in domestic investment
- c. a sudden increase in demand for the country's computer exports due to a faster chip design
- d. an increase in domestic prices caused by a labor shortage

Answers:

4. a. An increase in taxation will shift the IS* curve inward. To keep the exchange rate fixed, the Federal Reserve will shift in the LM* curve. Output will decrease. The trade balance will decrease because of the drop in savings due to the fall in output ($NX = S - I$).

b. A decrease in investment will shift in the IS* curve, and the LM* curve will shift in to keep the exchange rate fixed. Output will fall. The effect on the trade balance is unclear because both investment and savings have fallen. If savings falls by more than investment, then the trade balance will fall.

c. A sudden increase in demand for the country's computer exports will shift out the net exports schedule. This in turn will shift out the IS* curve. The LM* curve will shift out to keep exchange rates fixed, and output will increase. The trade balance will increase due to the increase in savings caused by higher output.

d. An increase in domestic prices will shift the LM* curve inward. This will put pressure on the exchange rate to increase, so the Federal Reserve will have to shift the LM* curve back out. The result is no change for output or the trade balance.

5. Explain how fiscal policy is ineffective under a flexible exchange rate.
- a. In previous chapters, we discussed how fiscal policy crowds out investment. Is that true for small open-economy countries? Why or why not?
 - b. What component of output is crowded out by government spending? What could the government or central bank do to prevent this crowding out?

Answers:

5. a. With flexible exchange rates, the output level is determined in the money market where the interest rate is set equal to the world interest rate. Any change in government policy will only affect the exchange rate.

b. In small open-economy countries, fiscal policy will not crowd out investment because the interest rate is fixed.

c. An increase in government spending will shift out the IS^* curve and drive up the exchange rate. This action will crowd out exports as they become more expensive for foreigners. The central bank could reduce the crowding out of exports by increasing the money supply and shifting out the LM^* curve.

6. Do you think the United States and Canada could benefit from a monetary union? Discuss the pros and cons of this proposal. Which country do you think would benefit most from such a union?

Answer:

6. The United States and Canada could benefit in several ways from monetary union. A common currency would eliminate the need to exchange currencies and reduce the risk associated with unexpected changes in exchange rates. One cost of a monetary union would be the loss of separate monetary policies. In addition, a central government would be needed to redistribute resources among regions.

The populations of each country are fairly mobile, so people may be willing to move to more prosperous regions to alleviate a regional recession.

Chapter 11

1. Assume that the economy is at full employment. The government decides to cut taxes to give the economy an extra boost.

a. Show the short run effect of this tax cut using the IS-LM model. What will happen to output and the interest rate?

b. What will happen in the long run?

c. If the Federal Reserve is following a policy of price stability, how should they react to the tax increase? If the Fed action is implemented, will the tax cut succeed in boosting output?

1. a. The tax cut will shift out the IS curve. Output and interest rates will rise.

b. In the long run, prices will rise due to the fact that output is above the full-employment level. This decreases the real money supply (M/P), and results in an inward shift of the LM curve.

c. If the Federal Reserve wanted to avoid the price increase, then it should decrease the money supply in the short run. This will shift in the LM curve and output will return to the full employment level. Thus, there was no boost to output.

2. In previous chapters, the government-purchases multiplier described the impact of a change in government spending on output. In the IS-LM model is multiplier effect larger or smaller than in the simple Keynesian-cross model? (Hint: Use the IS-LM graph to show how assumptions of each model lead to different multiplier effects.)

answer

2. In the IS-LM model, the government multiplier effect is smaller due to the crowding out of investment. An increase in government spending will shift out the IS curve and result in higher interest rates. The higher interest rates will lower investment spending, thereby lowering the boost to output. The difference in multipliers can be seen on the IS-LM graph. If the interest rate is held constant (by a flat LM curve), and increase in government spending will have a large effect

on output. If the LM curve is upward sloping, the output boost due to the government spending is diminished by the interest rate increase.

3. Why does the government spend money surveying consumers and firms on their current level of confidence in the economy? How might the Federal Reserve react to a sudden drop in consumer confidence?

3. Consumer and producer confidence surveys provide a good gauge for expectations. If people suddenly become pessimistic, they may decrease consumption. This will lead to a fall in output, and the consumer pessimism will have led to a downturn in the economy. The Federal Reserve, however, can react to confidence surveys by changing the money supply to prevent the undesirable movements in output.

4. Describe how the following events change the aggregate demand curve:

a. a decrease in money demand caused by the introduction of a new electronic money card

b. a decrease in the money supply

c. an increase in the price level

d. an increase in government taxes

4. a. A decrease in money demand will shift the LM curve outward. The aggregate demand curve should also shift outward because there will now be a higher output level at any given price.

b. A decrease in the money supply will shift the LM curve inward and decrease output. This translates into an inward shift of the aggregate demand curve.

c. An increase in the price level is displayed by a movement along the aggregate demand curve.

d. An increase in government taxes will shift in the IS curve and reduce output. The aggregate demand curve will shift in to reflect the lower output at any given price.

5. Explain the Pigou effect and why economists thought this would help end the Great Depression. Why did this theory fail to improve the economy? What actions should fiscal and monetary authorities have taken to end the depression sooner?

5. The Pigou effect is the prediction that falling prices will expand income due to the increase in real money balances (M/P). Consumers should feel wealthier since their money will now buy more goods than before. Therefore, they should increase spending. Unfortunately, the Pigou effect was dominated by other problems in the economy and output continued to fall. Some of these problems included debt deflation and the effects of a fall in price expectations. The Federal Reserve could have helped the situation by increasing the money supply to avoid the destabilizing deflation. Increased government spending would have reduced unemployment and helped increase aggregate demand.

Chapter 10

1. Explain how velocity is affected by interest rates and money demand. If the Federal Reserve increases the money supply, will there be a bigger change in output if the velocity remains constant or if it adjusts as you just described? (Hint: Use the quantity theory of money equation to discuss the impact of changes in money and velocity on output.)

Answer:

1. Interest rates affect the money demand of individuals. As interest rates increase, people hold less money because their opportunity cost of holding money (the interest they could receive in a savings account) has increased. A change in money demand will in turn affect the velocity. With a lower money demand caused by higher interest rates, the velocity of money will increase because the money held people will be used more frequently.

Using the quantity theory of money equation, $MV=PY$, we see that an increase in the money supply will lead to a proportional increase in output if velocity is constant (assuming prices are fixed). The increase in the money supply will lead to an increase in money demand and a decrease in interest rates, as shown in the money-market equilibrium graph. If velocity is a function of interest rates, we will see a decrease in velocity as people hold more money, and therefore spend the money as a slower rate. Using the quantity theory equation, the fall in velocity will diminish the increase in output caused by the increase in the money supply.

2. Let the following equations characterize an economy:

$$C = 400 + 0.8*(Y-T)$$

$$G = 300$$

$$T = 250$$

$$I = 200$$

- a. Draw a graph of planned expenditures for this economy and calculate the equilibrium level of output.
- b. Suppose output this year was 3000. Is the economy in equilibrium?
- c. If the government wanted to use fiscal policy to move the economy to equilibrium, by how much would it have to increase government spending? What is the government spending multiplier?
- d. If the government decided not to change fiscal policy, describe the process by which the economy would move toward equilibrium.

Answers:

2. a. The equilibrium is calculated by solving the following equation for output:

$$Y = C(Y-T) + I + G$$

$$Y = 400 + 0.8*(Y-250) + 200 + 300$$

$$Y = 900 + 0.8*Y - 0.8*250 = 700 + 0.8*Y$$

$$(1-0.8)Y = 700$$

$$Y = 700/(1-0.8) = 3500$$

In the graph, the planned expenditure line should have an intercept of 700 and a slope of 0.8. It should cross the $Y=E$ line at $Y=3500$.

- b. If actual expenditure is 3000, then the economy is below the equilibrium value.
- c. The government spending multiplier is $1/(1-MPC) = 5$. To increase output from 3000 to 3500, government spending should be increased by 100. The multiplier effect is $100*5 = 500$, which will increase output to the equilibrium level.

If the government does not increase spending, the economy will move toward equilibrium on its own. From the graph we see that actual output is less than planned expenditure. Therefore, firms are seeing an unplanned decrease in

inventories. They will boost production to meet the planned expenditures until the equilibrium level is reached.

3. Describe how the following changes will affect the LM curve

- a. An increase in the money supply
- b. An increase in output
- c. A one-time increase in the price level
- d. A decrease in the money demand function caused by the increased use of ATM machines (people demand less money at any given interest rate)

Answers:

3. a. An increase in the money supply (holding output constant) will lead to a lower interest rate, thereby shifting the LM curve downward.

b. An increase in output will lead to an increase in money demand and an increase in interest rates. This relationship is the basis for the LM curve, so we simply move along the curve.

c. A one-time increase in the price level will cause a decrease the real money supply, M/P , assuming the nominal money supply (M) remains fixed. This will cause an increase in interest rates and shift the LM curve upward.

d. A decrease in the demand for money caused by the increased use of ATM machines is illustrated by an inward shift of the money demand function in real money equilibrium graph. This will lead to a lower interest rate and will cause the LM curve to shift downward.

4. In previous chapters we discussed how increased government spending leads to crowding out of investment. Based upon the model in this chapter used to generate the IS curve, will an increase in government spending lead to a decrease in investment? Explain.

Answer:

4. An increase in government spending will lead to an increase in output at any given interest rate. Therefore, this is represented by an outward shift in the IS curve. If we now look at these two curves and hold the output level constant, we see that the shift has resulted in a higher interest rate. This would lead to a decrease in investment. Here the output level has not changed, so the increase of government spending has led to an equal decrease in investment — total crowding out. The extent of crowding out in equilibrium will be determined by the effect of government spending changes on the interest rate. If interest rates are unchanged, there is no crowding out, but as the interest rate increases, there will be a higher degree of crowding out.

5. Explain why the tax multiplier is smaller than the government-purchases multiplier.

Answer:

5. The tax multiplier is smaller than the government-purchases multiplier because of the initial effect on the economy. When the government spends an additional \$100, the entire amount contributes to output ($Y=C+I+G$). When taxes are reduced by \$100, the effect on the output depends on the change in consumption. If the marginal propensity to consume is 80%, that means only \$80 of the \$100 tax cut will be spent. The \$20 used for savings does not contribute to the output in the formula above. Thus, the government spending impact on the economy is $\$100/(1-0.8)$, whereas the tax cut contributes $\$100 \cdot 0.8/(1-0.8)$ or $\$80/(1-0.8)$.

Chapter 9

1. Explain why the long-run aggregate supply curve is vertical and the short-run aggregate supply curve is horizontal.

1. The short-run aggregate supply curve is horizontal because prices are assumed to be sticky in the short run. Therefore, producers will be willing to increase quantity supplied at a constant price. The long run aggregate supply

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curve is vertical because output is determined in the long run by the amount of labor and capital in the economy, along with the available technology. Therefore, changes in price will not affect the amount supplied.

2. Supposed there is an increase in the velocity of money cause by the increased use of ATM machines.

a. How would prices and output be affected in the short run?

b. If the Federal Reserve's objective is to keep prices stable in the short-run, how should it respond?

c. In the long run, what will happen to prices and output? If the Federal Reserve's objective is to keep prices stable in the long run, how should it respond? When should it undertake its response -- immediately or once the "long run" is reached?

d. Based upon your answer in part (c), how would you respond to those who argue that the Federal Reserve should wait until inflation appears before reacting?

Answers:

2. a. In the short run, prices remain fixed. According to the quantity theory of money, the increase in velocity must be matched by an increase in output since the money supply and prices are fixed.

b. Since prices are fixed in the short run, the Federal Reserve does not have make any response initially.

c. In the long run, output is determined by the factors of production. Therefore, the increase in velocity will not affect output. Prices must increase to match the higher velocity if output and the money supply remain unchanged. With a long-run goal of price stability, the Federal Reserve needs to decrease the money supply to offset the effect of the higher velocity. In terms of the quantity theory equation, $MV=PY$, if the decrease in money is proportional to the increase in velocity, there will be no change in the right-hand side of the equation. This monetary policy response should be enacted immediately so that there is no short run pressure on output. If output increases, this will lead to price increases in the long run to reduce output to its original position.

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d. Due to price stickiness in the short run, inflation pressures build up before inflation is actually present in the economy. To prevent the price increase, the Federal Reserve needs to respond to these pressures before the inflation occurs. If the Federal Reserve waits until inflation is visible, its response will be too late because changes in the money supply only have a long-run impact on prices.

3. The length of time between the short run and long run varies across sectors of the economy. For the following industries, estimate the time it takes to reach the "long run" based upon your knowledge of how quickly prices in this industry change in response to the given shock. Describe the rationale underlying each of your estimates.

- a.** gasoline prices after an announcement of a reduction in OPEC production
- b.** Sears catalog prices after an increase in the price of cotton used for making clothing
- c.** orange juice prices after a severe frost destroys crops in Florida
- d.** GM car prices after auto workers receive an increase in wages

Answers:

3. a. Gasoline prices respond very quickly to changes in oil production. The long run may be reached in as soon as 2 weeks.

b. Since catalogs are only printed a few times each year, it may be six months to a year before increases in input prices affect the price of clothing.

c. The price of oranges reacts very quickly to changes in production. A severe frost will probably translate into higher juice prices with a month.

d. Car prices may respond slowly to wage increases because prices are often established on a yearly basis that corresponds with the introduction of new models. The impact may be more indirect in terms of fewer rebate offers and fewer sales.

4. Suppose there was a huge drought in the U.S. that cut farm production in half for one year.
- a. What happens to the aggregate demand and aggregate supply curves?
 - b. How will output and prices change in the short and long run?
 - c. How would this affect net exports in the short run?
 - d. If the drought continued for several years, what would you expect to happen to net exports and the real exchange rate? (Hint: Use the models in Chapter 8 to describe the long-run changes. Assume that real interest rates are fixed.)

Answers:

4. a. In the short run, the aggregate supply curve would shift up. The aggregate demand curve and the long-run aggregate supply curve remain unchanged.
- b. In the short run, prices will increase and output will decrease. There will be no change in the long run.
- c. In the short run, U.S. agricultural exports should decrease due to the reduced production, and imports will increase as consumers buy food from other countries.
- d. In Chapter 8, we learned that net exports are determined by the difference between savings and investment. If there is no change in savings and investment, then net exports will not change. A decrease in demand for net exports caused by lower exports and higher imports would lead to a fall in the real exchange rate. If the fall in production was large enough to lead to a significant fall in U.S. GDP, this may lead to a fall in savings. If savings decreased (assuming no change in investment), then net exports would fall and the real exchange rate would rise, as seen in the real exchange rate equilibrium graph from Chapter 8.

Chapter 8

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1. The United States has run a trade deficit over the past two decades. Has this deficit been good or bad for the economy? Be sure to discuss the implied change in net foreign investment and the advantages/disadvantages of this change?

Answer:

1. Over the past two decades, the United States has imported more than it has exported. This has been a benefit in the sense that consumers have been able to consume a wider array of goods and increase the standard of living. The potential downside of the increase in imports over exports is that the U.S. has had to sell assets in order to pay for the excess imports, i.e. net foreign investment has decreased due to the trade deficit. If the sale of U.S. assets results in foreign investment in profitable ventures, the U.S. will benefit in terms of strong growth of the economy and low unemployment. If the trade deficit were to continue for several more decades, the concern would be over the size of foreign investment in the U.S. This could lead to problems seen in other countries when foreign investors suddenly lose faith in a country's economy. This loss of confidence can lead to a huge crisis as they quickly withdraw their investments.

2. In the 1980's, the United States had large government deficits and large trade deficits. In the late 1990's, the government deficit has been eliminated but the large trade deficit remains. Using the link between net exports, savings, and investment, why did the trade deficit remain so large when the government deficit was eliminated?

Answer:

2. The link between government deficits and trade deficits can be most easily seen by splitting savings into two categories: private savings (S^P) and government savings. Government savings is simply the difference between taxes (T) and government spending (G). So we can describe savings as $S = S^P + (T - G)$. Substitute this into the relationship between net exports, savings, and investment to get $NX = S^P + (T - G) - I$. Now we can see that if private savings and investment remain constant, a decrease in government savings (increased budget deficit) will lead to a decrease in net exports (increased trade deficit). This describes the situation in the 1980's. In the late 1990's we have seen an increase in government savings. This has not lead to a corresponding increase in net exports, which must indicate that private savings and investment have changed. Higher investment and lower private savings could account for this outcome, and there is evidence that both of these changes have occurred.

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3. Suppose that the expected inflation rate is 10 percent in the United States and 5 percent in Japan. The real interest rate is 3 percent in both countries, and assume that purchasing power parity holds.

- a. What is the nominal interest rate in each country?
- b. What are the expected changes in the real and nominal exchange rates?
- c. If the Federal Reserve increases the money supply and inflation expectations rose to 12 percent in the U.S., how would this change affect the expected changes in the real and nominal exchange rates? If you had planned to invest money in the United States, would this policy change affect your decision?

Answers:

3. a. The nominal interest rate is the sum of the real interest rate and expected inflation. For the U.S., this rate is 13 percent ($3+10$). For Japan, this rate is 8 percent ($3+5$).

b. Differences in inflation rates will lead to expected changes in the nominal exchange rate. The Yen/\$ rate should decrease by the differential in expected inflation, which is 5 percent ($5-10$). The real exchange rate is not expected to change unless there is a fundamental change to savings, investment, or demand for net exports.

c. An increase in inflation expectations in the U.S. would lead to a higher expected decrease in the Yen/\$ exchange rate. This expected decrease is now 7 percent ($5-12$). The real exchange rate should remain unchanged. For a U.S. investment, you would now need to receive a higher nominal interest rate or else your real return fall due to the increase in inflation.

4. Suppose the U.S. government has decided to enact a quota on Swiss watches to reduce the trade deficit. What impact will this quota have on the trade balance? How will this change affect real and nominal exchange rates? What impact does the quota have on U.S. importers and exporters who are not involved in trading watches?

Answer:

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4. The new quota will have no impact on net exports unless there is a change in savings or investment. The quota will cause the real exchange rate to increase due to the decreased demand for imports in the U.S. The nominal exchange rate should increase by the same percentage as the change in the real exchange rate. Even though the total amount of net exports remains unchanged, individual importers and exporters will be affected by the change in exchange rates. The higher exchange rate means that U.S. goods will be more expensive, thus exports will fall. Imports will now be cheaper, so they will increase. The changes in imports and exports will offset the fall in imports of watches so that total net exports remain unchanged.

5. You are presented with the following foreign exchange situation: you can trade dollars and pounds in London at a rate of 1.5 \$/£ and in New York at a rate of 1.6 \$/£ .

a. Show through an example if it is possible to profit from currency trading in New York and London?

b. Would you expect these rates to exist for long in the market place? Why or why not?

Answers:

5. a. Suppose you start with \$100. If you buy pounds in London, you will receive 66.67 pounds ($100/1.5$) for your \$100. Now return to New York and buy \$. You will receive \$106.67 (66.67×1.6) and you have just made a profit of \$6.67.

b. Now if everyone tried to make a profit in this manner, the high demand for pounds in London would drive up the value of the pound (an increase in the \$/£ exchange rate). The high demand for dollars in New York would drive up the value of the dollar (a decrease in the \$/£ exchange rate). This process would continue until the exchange rates are equal, at which point there is no opportunity to profit.